

Part B: Scan Your Local Machine

Based on your scan of localhost:

- **Total Open Ports:** 4
- **Port Numbers:** 135, 445, 3306, 8080
- **Services Running:** msrpc, microsoft-ds, mysql, http-proxy

Part C: Service Version Detection

Port	Service	Version
135	msrpc	Microsoft Windows RPC
445	microsoft-ds	(Version not specifically detailed, Windows default)
3306	mysql	MySQL 8.0.44
8080	http	Jetty 12.1.5

Part D: Operating System Detection

Was the operating system detected? *Yes*

Which OS was identified? *Windows 11*

Why is OS detection useful during penetration testing?

OS detection is critical because it allows penetration testers to identify the specific platform and version running on a target. This helps them narrow down potential vulnerabilities (CVEs) specific to that operating system, tailor their exploits, and understand the target's underlying architecture to plan the next steps of the attack.

Part E: Common Port Research

Port	Protocol	Purpose	Common Security Risks
20/21	FTP	Used for transferring files between a client and server.	Transmits data and credentials in cleartext, making it highly vulnerable to packet sniffing and interception.

22	SSH	Provides secure, encrypted remote command-line access.	Vulnerable to brute-force attacks if weak passwords are used, or compromised if private keys are leaked.
23	Telnet	An older protocol for remote command-line access.	Completely unencrypted. All commands and credentials are sent in cleartext, easily intercepted by attackers.
25	SMTP	Used for sending and routing email over the internet.	Often targeted for spam relaying, email spoofing, and Phishing if not secured properly.
53	DNS	Translates human-readable domain names into IP addresses.	Susceptible to DNS cache poisoning, spoofing, and DDoS amplification attacks.
80	HTTP	Used for transmitting web page data over the internet.	Unencrypted communication, allowing attackers to intercept data via Man-in-the-Middle (MitM) attacks.
110	POP3	Used by email clients to retrieve messages from a server.	Transmits email and passwords in cleartext unless wrapped in SSL/TLS.
143	IMAP	Used for retrieving and synchronizing email across multiple devices.	Similar to POP3, it is vulnerable to cleartext interception if not encrypted.
443	HTTPS	The secure, encrypted version of HTTP for web browsing.	Vulnerable if using outdated SSL/TLS certificates or weak encryption ciphers.
445	SMB	Used for file and printer sharing on Windows networks.	A frequent target for remote code execution exploits (e.g., EternalBlue) and ransomware propagation.
3389	RDP	Allows a user to remotely view and control a Windows desktop.	A prime target for brute-force attacks and ransomware deployment if exposed directly to the internet.

Part F: Scan Analysis

Write these in your own words in your README, but here is a guide based on your specific results:

1. **Which services are currently running?** My local machine is currently running Microsoft Windows RPC (Port 135), Microsoft-ds/SMB for file sharing (Port 445), a MySQL database server version 8.0.44 (Port 3306), and a Jetty HTTP web server version 12.1.5 (Port 8080).
2. **Are all open ports necessary?** Ports 135 and 445 are standard for Windows networking and file sharing, but ports 3306 and 8080 are only necessary when I am actively developing or testing database and web applications. If I am not currently coding, they do not need to be open.
3. **Which services could become security risks if misconfigured?** Port 445 (SMB) is a major security risk if exposed to the public internet, as it is frequently targeted by ransomware and exploits like EternalBlue. Additionally, Port 3306 (MySQL) could be a risk if it has weak or default administrative passwords.
4. **Which port would you disable if it wasn't required?** I would disable Port 8080 (Jetty HTTP) and Port 3306 (MySQL) when not actively working on development projects to reduce my system's attack surface.

Part G: Scan Report

Scan Date: June 30, 2026 **Target System:** Localhost (127.0.0.1) **Commands Used:**

- nmap --version
- nmap localhost
- nmap -sV localhost

Open Ports & Running Services:

- **Port 135 / TCP:** Microsoft Windows RPC
- **Port 445 / TCP:** Microsoft-ds (SMB)
- **Port 3306 / TCP:** MySQL 8.0.44
- **Port 8080 / TCP:** Jetty HTTP 12.1.5

Operating System: Windows

Observations: The scan completed very quickly (under 11 seconds) due to scanning locally. The system is exposing standard Windows networking ports alongside active development environments (MySQL and a Jetty web server).

Recommendations: Ensure that Windows Firewall is configured to block external traffic to port 445 to prevent SMB-based attacks. Shut down the MySQL and Jetty services when development tasks are completed to minimize unnecessary open ports. Ensure the MySQL root user has a strong password.

Conclusion: Through this scanning and enumeration task, I learned how to use Nmap to identify the active attack surface of a system. By discovering which ports are open and explicitly identifying the versions of the services running behind them, a penetration tester can map out potential vulnerabilities. I also learned that everyday development tools, like local databases or web servers, actively expose ports that could be leveraged by an attacker if left unpatched or misconfigured. This phase highlights the importance of the principle of least privilege—shutting down any service that isn't strictly necessary to reduce risk. Furthermore, performing this only on an authorized local machine reinforced the ethical boundaries required in professional security assessments.